

PRANEETH KUMAR JINUKALA

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SUMMARY

Embedded Systems Engineer with 3+ years of experience developing high-performance firmware for aerospace, industrial IoT, and smart automation systems. I specialize in C, Embedded C, and C++ development on ARM Cortex-M platforms such as STM32 and ESP32, with strong experience in FreeRTOS and bare-metal programming. I have built real-time systems with deterministic performance and integrated protocols like CAN/CAN-FD, Ethernet, SPI, I2C, and UART. My work includes hardware bring-up, device driver development, and deep debugging using JTAG, GDB, logic analyzers, and oscilloscopes. I have implemented secure boot, AES encryption, TLS communication, and secure OTA updates following MISRA C and ISO 26262 guidelines. I focus on low-power optimization, memory management, and RTOS task scheduling to improve system efficiency and reliability. I also have hands-on experience integrating MQTT-based cloud connectivity using AWS IoT Core for scalable IoT solutions. I am comfortable working across the full firmware lifecycle, delivering stable, secure, and production-ready embedded systems.

SKILLS

- **Programming Languages:** C, Embedded C, C++, Python
- **Embedded Development:** Bare-Metal Programming, FreeRTOS, Real-Time System Design, Device Drivers, Bootloaders
- **Microcontrollers:** ARM Cortex-M, STM32, ESP32
- **Communication Protocols:** SPI, I2C, UART, CAN, CAN-FD, LIN, Ethernet, TCP/IP, MQTT
- **Debugging & Validation:** JTAG, SWD, GDB, Logic Analyzer, Oscilloscope, In-Circuit Debugging (ICD), Unit Testing (Unity, Catch2)
- **Embedded Security:** Secure Boot, Secure OTA Updates, TLS/SSL, AES Encryption
- **Cloud & IoT Integration:** AWS IoT Core, Device Shadow
- **System Optimization:** Low-Power Design, Power Management, Memory Optimization, RTOS Task Scheduling
- **Automotive & Safety Standards:** ISO 26262, MISRA C, CAN/LIN Stack Integration
- **Tools & Version Control:** STM32CubeIDE, Keil, IAR Embedded Workbench, Git, GitHub, Jira

WORK EXPERIENCE

Honeywell

Nov 2025 – Present

Embedded Systems Engineer

- Architected scalable embedded firmware in C++ for ARM Cortex-M7 processors, enabling deterministic real-time performance for safety-critical aerospace applications.
- Integrated CAN-FD and Ethernet-based communication stacks with redundancy handling and fault-tolerant mechanisms, improving system reliability by 28% during hardware-in-loop validation cycles.
- Implemented secure boot chains, hardware-based key storage, and TLS-encrypted telemetry pipelines to align with aerospace cybersecurity and DO-178C compliance requirements.
- Led performance optimization efforts through RTOS task profiling, memory segmentation tuning, and low-latency interrupt handling, reducing firmware execution latency by 18% while maintaining MISRA C compliance and structured code reviews.

Honeywell

Feb 2025 – Oct 2025

Embedded Systems Engineer (Intern)

- Contributed to the development of modular firmware components in C for ARM Cortex-M platforms, enabling secure sensor data acquisition and reliable gateway communication within smart building automation environments.
- Designed and validated FreeRTOS-based multitasking architecture to manage HVAC control signals and real-time diagnostics, improving task scheduling efficiency by 20%.
- Implemented secure MQTT and TLS-based communication between edge devices and cloud dashboards, aligning with Honeywell's cybersecurity standards for industrial IoT deployments.
- Performed hardware bring-up and peripheral integration using SPI, I2C, and UART protocols, leveraging oscilloscopes and logic analyzers to resolve signal timing and memory-mapped I/O issues.
- Automated regression testing using Python-based scripts and Unity framework, reducing manual validation effort by 15% and enhancing firmware release stability.
- Documented system architecture, safety compliance considerations, and version-controlled releases through Git and Jira workflows, supporting scalable deployment practices aligned with U.S. Industry 4.0 and smart infrastructure initiatives.

TatvaSoft

Jan 2021 – Jul 2023

Embedded Systems Engineer

- Engineered firmware for STM32-based edge devices using Embedded C and integrated MQTT-based cloud communication to enable secure, real-time telemetry streaming for industrial IoT environments.

- Developed bare-metal and FreeRTOS-based task scheduling modules to manage sensor acquisition, event handling, and peripheral control, improving firmware response efficiency by 30%.
- Implemented SPI, I2C, and UART drivers for multi-sensor interfacing, ensuring stable data exchange and hardware abstraction across custom ARM Cortex-M boards.
- Optimized low-power modes and dynamic clock scaling techniques for battery-operated IoT prototypes, extending device operational life by 25% during field validation.
- Configured secure bootloaders with AES-based firmware encryption and OTA update support, strengthening embedded security compliance for connected edge systems.
- Executed board-level debugging using JTAG, SWD, logic analyzers, and oscilloscopes to troubleshoot signal integrity and memory faults, reducing hardware-software integration defects by 20%.
- Collaborated with cross-functional teams to integrate CAN and TCP/IP stacks for industrial gateway communication, aligning firmware architecture with scalable Industry 4.0 solutions.
- Contributed to entry-level U.S. trending projects including smart home automation prototypes and wearable health monitoring devices, applying MISRA C guidelines, Git-based version control, and structured unit testing for ATS-aligned embedded development practices.

PROJECTS

Low-Power IoT Environmental Sensor Node

- Designed and developed an ESP32-based IoT sensor node using Embedded C, implementing deep-sleep mode and advanced power management techniques for energy-efficient edge operation.
- Integrated secure MQTT communication over TCP/IP with AWS IoT Core, enabling real-time cloud-based data monitoring and remote device management.
- Optimized system performance through memory management, clock scaling, and low-power firmware design, significantly reducing overall power consumption and improving device battery life.

Real-Time Motor Control System with Ethernet Connectivity

- Designed and developed high-performance embedded firmware in C/C++ for ARM Cortex-M microcontrollers, leveraging PWM, ADC, and DMA peripherals to implement deterministic, closed-loop motor control algorithms for industrial automation applications.
- Implemented robust Ethernet-based TCP/IP communication stack to enable real-time data acquisition, remote monitoring, and supervisory control integration with SCADA and HMI systems.
- Performed in-depth performance profiling, latency analysis, and real-time debugging using GDB, JTAG, and In-Circuit Debugging (ICD) tools, optimizing interrupt handling and memory usage to ensure low-latency and reliable execution under real-time constraints.

EDUCATION

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| • Master of Science in Computer Engineering
California State University, Northridge, USA GPA: 3.21 | Aug 2023 – Dec 2025 |
| • Bachelor of Engineering in Electrical and Electronics Engineering
C.V.S. R College of Engineering, Hyderabad GPA: 6.94 | Jul 2018 – May 2022 |